CLASS 10 — SCIENCE · FORMULA SHEET

Chapter 4: **Carbon and its Compounds** | Quick revision, one page

Bonding in carbon

- Carbon: atomic number 6 (2, 4), tetravalent; forms covalent bonds by sharing electrons (does not form C^{4+} or C^{4-}).
- Covalent compounds: low melting/boiling points, poor conductors. Catenation: self-linking into chains, branches, rings.
- Allotropes: diamond (hard, non-conductor), graphite (soft, conducts), fullerene C_{60} (football-shaped).

Hydrocarbons and homologous series

- Alkanes C_nH_{2n+2} (single bonds): methane, ethane, propane, butane, pentane.
- Alkenes C_nH_{2n} (double bond): ethene C_2H_4; alkynes C_nH_{2n-2} (triple bond): ethyne C_2H_2.
- Covalent bonds in an alkane $= 3n + 1$. Consecutive homologues differ by CH_2 (14 u).
- Functional groups: halo ($-$Cl, $-$Br), alcohol ($-$OH, -ol), aldehyde ($-$CHO, -al), ketone ($>$C=O, -one), carboxylic acid ($-$COOH, -oic acid).

Chemical properties

- Combustion: $CH_4 + 2\,O_2 \longrightarrow CO_2 + 2\,H_2O$ + heat and light; saturated burn with clean blue flame, unsaturated with yellow sooty flame.
- Oxidation: alcohol $\xrightarrow{\text{alk. } KMnO_4 \text{ or acidified } K_2Cr_2O_7}$ carboxylic acid.
- Addition (unsaturated, Ni catalyst): $CH_2{=}CH_2 + H_2 \longrightarrow CH_3{-}CH_3$. Substitution (saturated, sunlight): $CH_4 + Cl_2 \longrightarrow CH_3Cl + HCl$.

Ethanol and ethanoic acid

- $2\,Na + 2\,C_2H_5OH \longrightarrow 2\,C_2H_5ONa + H_2$; dehydration: $CH_3CH_2OH \xrightarrow{\text{conc} \cdot H_2SO_4,\ 443\,K} CH_2{=}CH_2 + H_2O$.
- $CH_3COOH + NaOH \longrightarrow CH_3COONa + H_2O$;
  $2\,CH_3COOH + Na_2CO_3 \longrightarrow 2\,CH_3COONa + H_2O + CO_2$.
- Esterification: acid + alcohol $\xrightarrow{\text{conc. } H_2SO_4}$ ester + water. Saponification: ester + NaOH $\rightarrow$ alcohol + sodium salt of acid (soap).
- Vinegar: 5-8% ethanoic acid; pure ethanoic acid freezes at 290 K (glacial acetic acid).

Soaps and detergents

- Soap: sodium/potassium salt of long-chain fatty acid; ionic head (hydrophilic) + hydrocarbon tail (hydrophobic) $\rightarrow$ micelles trap oily dirt.
- Hard water (Ca^{2+}, Mg^{2+}) + soap $\rightarrow$ insoluble scum; detergents (sulphonate / ammonium salts) do not form scum.